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Zhenghe Guo

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ABOUT ME

Final-year undergraduate researcher working on embodied systems and spatial intelligence, with a focus on learned state representations, world modeling, and planning.

EDUCATION AND TRAINING

2023/08/31 – 2027/06/01 Hangzhou, China

B.ENG. IN AGRICULTURAL ENGINEERING; MINOR IN ADVANCED CLASS OF ENGINEERING EDUCATION

Zhejiang University

Selected coursework (scores /100): Fundamentals of Artificial Intelligence (95), Automatic Control Theory (94), Cybernetics (94), Bioproduction Robotics (94), Probability and Mathematical Statistics (93), Frontiers of Artificial Intelligence (93), Theoretical Mechanics (93), Heat Transfer and Fluid Mechanics (92), Robotics Technology and Practice (91), Fundamentals of Mechanical Manufacturing (91), Mechanics of Materials (90).

Field of study Engineering, manufacturing and construction | **Final grade** 3.86/4.00 | **Level in EQF** EQF level 6 |

National classification 6 | **Type of credits** Chinese Higher Education Credits | **Number of credits** 187.0 |

Thesis Design of Motion Control Algorithm for Magnetically Actuated Microrobots Based on Reinforcement Learning

2025/06/29 – 2025/07/19 Kent Ridge, Singapore

VISITING STUDENT IN STATISTICS AND DATA SCIENCE National University of Singapore

Advanced Machine Learning & Tree-based Modeling: Completed intensive coursework at NUS focusing on decision tree algorithms (GUIDE, CART, CTREE, Random Forests). Implemented rigorous statistical tests (Chi-squared tests, Wilson-Hilferty approximation) to overcome variable selection bias and evaluated bias-adjusted feature importance scores.
Data Visualization & Exploratory Analysis: Leveraged Python (Jupyter Notebook, Pandas, Seaborn) and Tableau to build interactive dashboards and conduct exploratory data analysis on credit risk datasets, translating complex multi-variable distributions into actionable insights.

Field of study Information and Communication Technologies | **Final grade** 4.00/4.00

WORK EXPERIENCE

UNIVERSITY RESEARCH ASSISTANT – AGILE ROBOTIC TELE-SYSTEMS(ARTS) LAB,COLLEGE OF CONTROL SCIENCE AND ENGINEERING,ZHEJIANG UNIVERSITY – 2024/10/22 – Current – HANGZHOU, CHINA

Developed real-time stress-field reconstruction for layered-cast soft-tissue models, connecting sparse pressure measurements with curved-surface tactile sensing; optimized sensor layouts and used FEBio finite-element priors with Gaussian-mixture conditioning and posterior reweighting under uncertain material and loading conditions.

UNIVERSITY RESEARCH ASSISTANT – ROBOTIC MICRO/NANO MANIPULATION LAB, ZHEJIANG UNIVERSITY – 2025/02/01 – Current – HANGZHOU, CHINA

Developed hybrid physics–neural models of current-to-field mapping and actuation-to-motion response for magnetic microrobots; investigated Dreamer-based 3D navigation and PPO-based planar navigation with Transformer history encoding, contrastive context learning, and FiLM policy conditioning.

UNIVERSITY RESEARCH ASSISTANT – 3DV LAB, ZJU CAD&CG STATE KEY LABORATORY,ZHEJIANG UNIVERSITY – 2025/10/01 – 2026/08/14 – HANGZHOU, CHINA

Department: School of Software Technology, Zhejiang University | **Website:** https://github.com/zju3dv

Implemented dynamic street-scene reconstruction with COLMAP/FreeTimeGS and egocentric instance reconstruction with EFM3D/ShapeR; evaluated image-to-3D generation and articulated-object modeling

VISITING RESEARCH INTERN – TSINGHUA UNIVERSITY, INSTITUTE FOR AI INDUSTRY RESEARCH (AIR) – 2026/01/01 – 2026/02/14 – BEIJING, CHINA

Developed StereoHO-based human-to-robot handover with intent analysis, grasp planning around human-held regions, and stable transfer informed by object appearance and contents; implemented hand-eye calibration and robot execution.

VISITING RESEARCH INTERN – INCEPTION 3D LAB, WESTLAKE UNIVERSITY – 2026/05/10 – 2026/10/25 – HANGZHOU, CHINA

Investigated physics-constrained deformable dynamics, from recurrent state evolution to explicit MPM with graph-based residual correction; compared state representations, calibration, and correction interfaces for multi-step prediction. Authored a survey of 200+ sources on world-model validity and decision utility in deformable-object manipulation.

HONOURS AND AWARDS

2026/05/13

RoboCup China Open, Humanoid League (Small) – Chinese Association of Automation(CAA)

National First Prize, Runner-up

2025 The 3rd Quantum Information Technology and Application Innovation Contest – China Academy of Information and Communications Technology(CAICT)

National First Prize,Runner-Up (Mathematical Modeling Innovation Contest Based on Coherent Optical Quantum Computers)
– Project: “Coherent Ising Machine-based Decoding Optimization for LDPC Codes in Wireless Communication Networks”

2025/11/30

2025 Asia and Pacific Mathematical Contest in Modeling (APMCM) – Beijing Society of Image and Graphics

National First Prize (motion planning framework for Unitree G1 humanoid robot dance performance, top 2%)

2026 China International College Students’ Innovation Contest

Gold Medal, project: Flexskin: a Lightweight Custom Functional Skin for Embodied AI

Scholarships

Xiangyang Scholarship, First-Class(RMB 10,000,sole annual recipient in the college)
Keyue Zhongkai Scholarship

2024/12/01

The 16th National Undergraduate Mathematics Competition – Chinese Mathematical Society(CMS)

Second Prize(Non-mathematics A category)

PUBLICATIONS

2026

ObjectTwin: Measuring the Asset Correspondence Gap for Capture-Anchored Digital Twins.

Instance-level reconstruction from RGB-D captures, connecting segmentation, VLM filtering, generative asset completion, scan-guided composition, and pose refinement. The work evaluates when a visually convincing asset remains geometrically mismatched to its captured object.

Authors: Zhenghe Guo

Link https://styokuok.github.io/sim_ready/

2026

Spatio-Temporal Transformer Meta-RL for Robust Magnetic Microrobot Navigation in Cluttered Environments.

Context-aware Meta-RL for robust autonomous navigation of magnetic microrobots in cluttered environments, using spatio-temporal observations to adapt control under changing physical context and partial observability.

Authors: Fanghao Wang* , Zhenghe Guo* , Yihao Hu* , et al.

2026

Sample-Efficient Residual Dynamics for Contact-Mismatched Deformable Simulation

Learning residual corrections under a controlled contact-model discrepancy. Short-rollout training is evaluated through complete-episode prediction and candidate-action comparisons, testing when learned corrections transfer and improve downstream decisions.

Authors: Zhenghe Guo

Link https://styokuok.github.io/res_dyn/

2026

Beyond State Prediction: Learning Action Consequences in Visual Physical Systems

Restoring the same physical state for each candidate action produces complete response tables. Fluid experiments separate information about the current state, prediction of action consequences, and the realized utility of the selected action, including tests on unseen actions and hidden motion.

Authors: Zhenghe Guo

Link https://styokuok.github.io/vis_phys/

2025

Rate Maximization for UAV-assisted ISAC System with Fluid Antennas

Joint optimization of UAV placement and fluid-antenna configuration for an integrated sensing and communication link, formulated to improve achievable rate under coupled sensing and communication constraints.

Authors: X. Yang, Z. Guo, S. Liang, Z. Yang, C. Zhu, Z. Zhang | **Journal Name**: IEEE/CIC ICCC Workshops 2025

PATENTS

Orthodontic Periodontal Ligament Stress Measurement Device and Stress Field Reconstruction Method

Xiaoyan Chen, Xinyi Fang, Gaofeng Li, Qianming Chen, Zhichao Fan, **Zhenghe Guo**, Peng Deng, Misi Si, Rui Zhang. Chinese invention patent application. *Pending*

Method for Forming and Assembling a Simulated Periodontal Ligament Layer and Associated Testing Device.

Gaofeng Li, Zhichao Fan, Xiaoyan Chen, **Zhenghe Guo**, Xinyi Fang, Peng Cheng, Jiming Chen.Chinese invention patent application. *Pending*.

Physical Field Reconstruction Method and Device.

Gaofeng Li, **Zhenghe Guo** (first student inventor), Xiaoyan Chen, Zhichao Fan, Xinyi Fang, Peng Cheng, Jiming Chen.Chinese invention patent application. *Pending*

PROJECTS

2025/11/02 – 2025/11/29
DynaMo: Contact-Implicit Dynamic Motion Optimization and Energy-Aware Control for Humanoid Choreography

Designed a dynamics-aware framework combining contact scheduling, ZMP-aware centroidal control, whole-body coordination, and electrical-energy analysis; Asia and Pacific Mathematical Contest in Modeling (APMCM) National First Prize (top 2%).

Link https://styokuok.github.io/dynamo/

2024/09/15 – 2026/05/30
ZJUDancer: Vision & Localization for KidSize RoboCup Soccer

Vision-group development for autonomous soccer: ball/field/robot perception, self-localization, odometry integration, and on-robot debugging under changing field conditions.

Link https://zjudancer.github.io/

2026/03/01 – 2026/07/04
RoboARM – Vision-Guided Gomoku Manipulation

The RoboARM system autonomously plays Gomoku using YOLO/ArUco board perception, Negamax move selection, ROS/HTTP communication, and a calibrated six-axis arm with a pneumatic end effector.

Link https://gitee.com/zjuatri/chess/tree/main/gobang_ws/src

2025/09/15 – 2026/01/04
TurtleBot Autonomous Navigation

The TurtleBot stack integrates Nav2 with a custom ROS RRT* planner, using goal-biased sampling, KD-tree collision checks, obstacle inflation, rewiring, and real-robot path execution.

Link https://styokuok.github.io/main/projects/turtlebot-navigation/

2025/11/01 – 2025/11/30
QUBO-Based Hybrid LDPC Decoding for Coherent Optical Optimization

Engineered an LDPC simulation and decoding prototype with belief-propagation baselines, QUBO construction, adaptive penalties, partitioned solving, and BER/FER/runtime evaluation.

Link https://styokuok.github.io/main/projects/quantum-ldpc/

CERTIFICATIONS

Nvidia, 2024/07/10
CUDA C/C++

NETWORKS AND MEMBERSHIPS

2023/10/01 – 2024/08/01 Hangzhou
Zhejiang University Student Robotics Association

Member (2023.10-2024.6), Vice President(2024.6-2024.9)
– Practiced designing and coding on wheeled robots, and engaged in frequent discussions with fellow members about tech topics such as compile languages, reinforcement learning for robots and so on.

2024/08/01 – 2025/01/01 Hangzhou
Zhejiang University ARC Agricultural Robotics Club

Member (2024.8-2025.1)
– Practiced robotics designing and coding skills in designing seeding robots and spraying robots.

Hangzhou
Team ZJUDancer (RoboCup2025 & 2026), Zhejiang University

Member of Computer Vision Group
– Continuing advances in vision and detection with real legged robots in an adversarail context.

SKILLS

Programming & Software:

Tableau (data analysis): Base | Windows, GNU/Linux, MacOS | C++ | R | MATLAB (including toolboxes: Deep Learning, Statistics and Machine Learning and Signal Processing) | Git/Github, Docker, Gitlab | Python (NumPy, Pandas, NLTK, Matplotlib, sklearn, TensorFlow) | Web development experience: React, Angular, JavaScript, TypeScript, HTML, CSS [frontend]

Machine Learning / AI frameworks:

image recognition | computer science | probability theory | Gaussian Processes for Machine Learning Toolbox | Deep Learning, Reinforcement Learning, Regression, SVM, RF, Naive Bayes, Clustering, LASSO | algorithms | statistics | PyTorch, Keras

Robotics

control methods | embedded systems | Artificial Intelligence and Automation in robotics | sensor fusion | kinematics, dynamics | path planning | Robotics Tools: ROS, OpenCV, PyGlet, PyBullet, PyChrono, OpenAI Gym, Docker | Computer Aided Design/ Computer Aided Manufacturing Applications | mechanical engineering | Arduino STM32 RPi | computer vision

Numerical & Scientific Computing

apply statistical analysis techniques | Parallel computing in HPC environment | mathematical modelling | Math - univariate and multivariate calculus, analysis, linear algebra, probability, optimization | Monte Carlo simulation

Soft Skills

Project management and planning | Team collaboration & teamwork | Problem-solving & independent research | Adaptability & continuous learning | Technical report writing & academic paper writing

LANGUAGE SKILLS

Mother tongue(s): **CHINESE**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C1	C2
SPANISH	A2	A2	A1	A1	A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HOBBIES AND INTERESTS

Hobbies

Long-distance running, music, and learning foreign languages

Social Activities

5 **UNESCO** course certificates, 1 **UNICEF** donation certificate for supporting Education in the Mountainous Region, volunteer in Hangzhou West Railway Station, and more than three years working experience as a Mental Health Representative